Highlights

Exploring the Relationship Between Mental Health Support Services and Academic Performance in Higher Education: A Leakage-Audited Analysis

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- Leakage auditing materially lowers apparent benchmark strength on both public student datasets.
- SHARE organizes clinical, service, and contextual variables through a service-aware analytical framework.
- On MHP-Multi, the strongest audited baseline reaches $F_1 = 0.902$, while SHARE reaches 0.825.
- On SMH-Kaggle, only six academic-risk positives remain, making all methods unstable after audit.
- Oracle analysis shows the current public service proxy adds little beyond stress-item information alone.

Exploring the Relationship Between Mental Health Support Services and Academic Performance in Higher Education: A Leakage-Audited Analysis

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Abstract

In higher education, mental health support services are often expected to help students sustain academic performance, but public datasets provide only limited evidence about how support-access variables relate to academic outcomes once direct label leakage is removed. We study this relationship using two public student datasets and a leakage-audited analytical framework, **SHARE** (**S**tudent **H**ealth-**A**cademic **R**elationship **E**ncoder), which organizes clinical, service, and contextual variables, and we compare it against strong generic baselines. On MHP-Multi, we examine whether stress, demographic, and support-access covariates can identify PHQ/GAD-defined risk after removing all PHQ-9 and GAD-7 totals and item responses from the input. On SMH-Kaggle, we examine whether symptom, treatment-seeking, and demographic variables are associated with academic risk (CGPA < 2.5) when CGPA itself is excluded from the predictors. Under these audited definitions, MHP-Multi remains informative but the empirical ceiling is moderate: the best baseline reaches $F_1 = 0.902$, while SHARE reaches $F_1 = 0.825$. On SMH-Kaggle, only six positive academic-risk cases remain, all methods become unstable, and the strongest model reaches $F_1 = 0.241$ while SHARE reaches 0.132. Follow-up diagnostics show that the full SHARE configuration attains the best cross-validated F_1 among its own ablations on MHP-Multi, but strong generic baselines still remain ahead, and oracle feature-set analysis indicates that the current service proxy adds little measurable gain beyond stress items alone. A brief GPA regression check is also weak, with simple Ridge baselines outperforming SHARE on both datasets. These findings suggest that current public higher-education datasets support only cautious, correlational conclusions about the relationship between mental health support services and academic performance. Stronger claims will require richer longitudinal service-utilization records, larger leak-free cohorts, and study designs that can better separate association from causation.

Keywords: higher education, mental health support services, academic performance, academic risk, student wellbeing, leakage-audited analysis

1. Introduction

Mental health support services are now part of the core student-success infrastructure in higher education because psychological distress is closely associated with lower academic performance, reduced persistence, and dropout risk (Eisenberg et al., 2009; Auerbach et al., 2018). Universities have expanded counseling and other support pathways, and service utilization has increased over time (Lipson et al., 2019). Prior work further suggests that counseling and broader social support can be associated with retention, wellbeing, and academic performance (Lee et al., 2009; Cholewa and Ramaswami, 2015; Ruihua et al., 2025). Yet the empirical relationship between support-service indicators and academic outcomes remains difficult to study with public datasets. Available student mental-health benchmarks often mix symptom variables, coarse support-access measures, and outcome labels in ways that can blur whether a model is learning a substantive service-performance association or merely reconstructing the label from overlapping inputs.

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Existing work only partially resolves this problem. One line of research predicts academic outcomes such as dropout or low performance from demographic, academic, and institutional covariates using conventional tabular models (Sghir et al., 2023; Alhardi and ALAN, 2024). A second line predicts mental-health states from survey responses or health-behavior signals (Morr et al., 2024; Abdul Rahman et al., 2023). A third line imports generic deep tabular architectures, including Transformer-style models and recent small-data learners, into educational and health-informatics settings (Gorishniy et al., 2021; Hollmann et al., 2024). These directions are useful, but they leave open the substantive question that motivates this paper: what can current public data actually tell us about the relationship between mental health support services and academic performance in higher education once direct label proxies are removed?

- **Fragile task definitions.** Public benchmarks can entangle labels with inputs, as when PHQ/GAD-derived labels are predicted from PHQ/GAD variables or academic-risk labels are predicted from CGPA itself. Such setups inflate observed performance and distort any inferred relationship between support-related covariates and student outcomes.
- **Coarse measurement of service engagement.** Public student datasets usually encode counseling, treatment-seeking, or support access as very limited indicators. This makes it hard to tell whether apparent signal reflects meaningful support-service association or simply correlation with broader stress and demographic profiles.
- **Unclear separation between data signal and model gain.** On small and imbalanced public datasets, performance ceilings may be governed more by residual feature informativeness than by architecture. Without an audit-first evaluation, stronger scores can be misread as evidence about service-academic relationships when they mainly reflect permissive task construction.

Guided by these observations, we use SHARE, a service-aware architecture that groups heterogeneous student records into clinical, service, and contextual tokens, as a structured framework for leak-free analysis of student support data. The main contributions of this study are as follows:

- **Relationship-focused, leakage-audited task reformulation.** We study two public student mental-health datasets through the lens of higher-education support services and academic outcomes, and we redefine their tasks to remove direct label-feature overlap. This reframing materially changes the empirical interpretation by shifting attention from permissive benchmark formulations to the audited performance ceiling of each dataset.
- **Service-aware analytical framework.** We apply SHARE as a compact framework for organizing clinical, service, and contextual inputs, combining semantic feature grouping, service-anchored attention, and joint classification-regression learning. This lets us examine whether service-related variables retain incremental association with student risk and GPA once obvious leakage is removed.
- **Evidence that current public service indicators are limited.** Across strong classical, tree-based, and modern tabular baselines, together with ablation and oracle-style analyses, we show that leak-free public benchmarks support only cautious conclusions about support-service and academic-performance relationships. In particular, the current MHP-Multi service proxy adds little beyond stress items, while SMH-Kaggle is too small for stable inference about academic risk.

2. Background and Related Work

2.1. Academic Performance, Retention, and Support Services

Studies in higher education have long reported links between mental health, counseling use, persistence, and academic success. Foundational work shows that mental health difficulties are associated with lower academic performance (Eisenberg et al., 2009), while counseling utilization and broader support systems have been linked to retention, wellbeing, and selected academic outcomes in institutional cohorts (Lipson et al., 2019; Lee et al., 2009; Cholewa and Ramaswami, 2015; Ruihua et al., 2025). This literature makes the substantive question clear: support services matter, but observed associations are embedded in selection processes because students who seek help often differ systematically from those who do not.

Alongside this substantive literature, predictive modeling for student outcomes has traditionally been framed as supervised learning on structured educational records, where the main goal is to estimate dropout, failure, or broader academic risk from demographic, behavioral, and administrative covariates. Early studies relied on conventional classifiers such as logistic regression, naïve Bayes, and decision-tree variants, while more recent work has shifted toward ensemble methods, especially Random Forest, XGBoost, and LightGBM, because they handle heterogeneous tabular inputs with limited preprocessing (Alyahyan and Düştegör, 2024; Jawad et al., 2022; Chen and Guestrin, 2016; Sghir et al., 2023; Alhardi and ALAN, 2024). These methods are practically useful, but they usually treat service variables as one feature block among many and rarely ask whether the benchmark itself is valid for studying support-service relationships. For the present problem setting, that is a meaningful limitation.

2.2. Mental Health Screening and Student Wellbeing Modeling

A parallel literature studies student mental health more directly by predicting depression, anxiety, stress, or well-being from questionnaires and health-related behaviors. Representative examples include machine learning models for university-student depression, anxiety, and stress screening (Morr et al., 2024), as well as behavior-based predictors built from lifestyle or self-reported health signals (Abdul Rahman et al., 2023; Chowdhury et al., 2025). The core modeling idea in this line is different from academic-performance modeling: instead of inferring downstream educational outcomes from institutional covariates, these methods seek clinically or behaviorally meaningful representations that are more proximal to psychological state.

This literature is valuable because it brings domain-relevant inputs into the modeling loop and shows that meaningful signal can be extracted from survey instruments or behavior traces. At the same time, its dominant task formulations differ from the one studied here. Many models predict mental-health state itself rather than the relationship between support-service indicators and academic outcomes, and they rarely treat service utilization as a structured source of evidence distinct from symptom burden. More importantly for this paper, when labels are derived from the same questionnaires that also populate the feature space, the line between clinically relevant analysis and direct label reconstruction can become blurred. These concerns motivate stricter task definitions before any substantive conclusion about support-service association is drawn.

2.3. Analytical Models for Heterogeneous Student Data

Recent work on deep learning for tabular data emerged partly because classical ensembles, although strong, offer limited control over how heterogeneous feature groups interact. TabNet introduced sequential attentive feature selection (Arik and Pfister, 2021); TabTransformer contextualized categorical features with Transformer blocks (Huang et al., 2020); and FT-Transformer showed that full feature tokenization with self-attention can be a competitive default for mixed tabular inputs (Gorishniy et al., 2021). SAINT further extended this family with row-wise attention and contrastive pretraining (Somepalli et al., 2021). More recent models have raised the baseline again: TabM emphasizes parameter-efficient ensembling for tabular deep learning (Gorishniy et al., 2024), and TabPFN v2 targets small-data tabular prediction through meta-trained in-context inference (Hollmann et al., 2024). In parallel, broader benchmark analyses continue to show that tree-based learners remain difficult to beat on many tabular problems (Grinsztajn et al., 2022).

The advance of this literature is not just architectural variety but a more demanding empirical standard: modern tabular baselines now capture high-order feature interactions while remaining competitive in exactly the low-sample regime that characterizes many public educational and mental-health datasets. Yet these models are intentionally general-purpose. They usually treat feature interactions symmetrically, assume that the benchmark task has already been specified appropriately, and do not explicitly distinguish service utilization from symptom or demographic descriptors. For the audited benchmarks considered here, that matters in two ways: strong generic models may already absorb most of the usable association, and any additional domain-specific inductive bias must justify itself after direct label proxies have been removed. This is the setting in which the present paper positions its analytical framework.

Taken together, prior work offers evidence that mental health, support services, and academic outcomes are linked, along with strong tools for modeling educational and clinical tabular data. What remains underdeveloped is a careful, leakage-aware analysis of how much current public datasets can actually say about the relationship between support-service indicators and academic performance. Our work places SHARE in that gap: not as a claim that a domain-specific architecture automatically dominates generic baselines, but as a service-aware framework for examining clinical, service, and contextual evidence once benchmark construction is made more defensible.

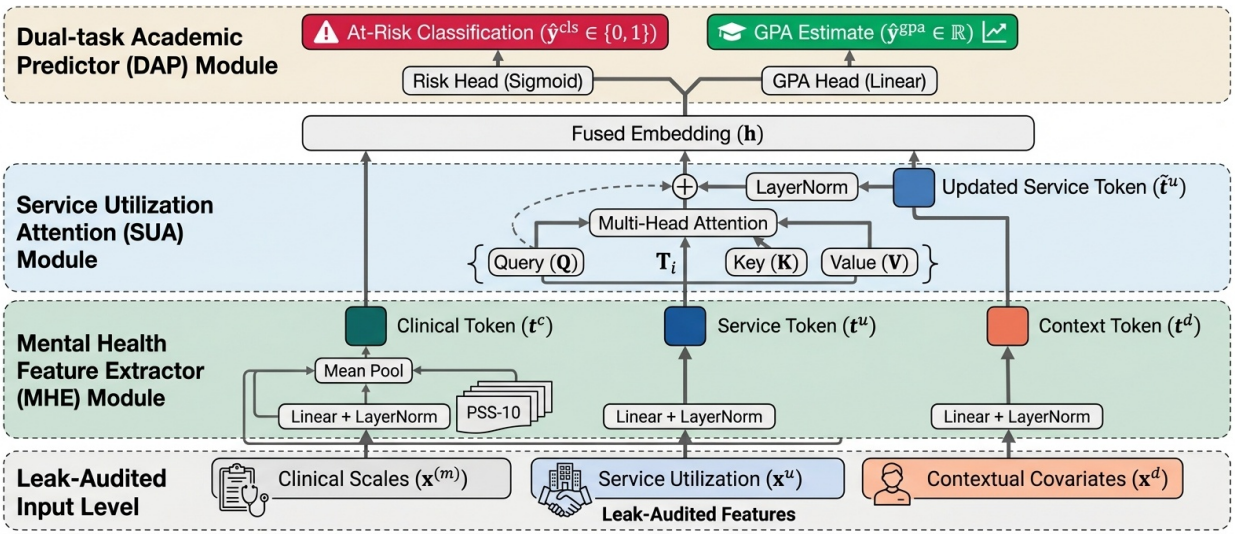

Figure 1: Overall architecture of SHARE. The schematic highlights the leak-audited input groups, the MHE encoder, the service-anchored SUA fusion, and the dual-task prediction heads used for risk classification and GPA analysis.

3. Materials and Methods

3.1. Overview

We introduce SHARE (**S**tudent **H**ealth-**A**cademic **R**elationship **E**ncoder) as a structured analytical model for examining higher-education support-service, mental-health, and academic-risk data with heterogeneous clinical, service, and demographic inputs. Figure 1 presents the overall framework. It consists of three key components:

- **Mental Health Feature Extractor (MHE):** Converts raw student records into a small set of semantic tokens representing clinical burden, service utilization, and contextual covariates, thereby reducing scale mismatch across instruments.
- **Service Utilization Attention (SUA):** Uses the service token as an anchor query to selectively weight clinical and contextual evidence when estimating how service engagement aligns with academic risk.
- **Dual-task Academic Predictor (DAP):** Simultaneously optimizes at-risk classification and GPA regression from the shared service-aware representation.

Given a student feature vector $\mathbf{x} \in \mathbb{R}^F$ comprising clinical scale scores, service utilization indicators, and contextual covariates, SHARE first constructs a compact token set for that student, then produces an at-risk score $\hat{y} \in \{0, 1\}$ and a GPA estimate $\hat{g} \in \mathbb{R}$ for downstream relationship analysis. The overall process is summarized in Algorithm 1.

3.2. Problem Formulation

We formulate leakage-audited analysis of support-service and academic outcomes as supervised learning on

$$\mathcal{D} = \{(\mathbf{x}_i, y_i^{\text{cls}}, y_i^{\text{gpa}})\}_{i=1}^N, \tag{1}$$

where $\mathbf{x}_i \in \mathbb{R}^F$ contains only the clinical, service-utilization, and contextual covariates that remain after removing direct label proxies; $y_i^{\text{cls}} \in \{0, 1\}$ is the primary binary at-risk label; and $y_i^{\text{gpa}} \in [0, 4]$ is the student's GPA (or equivalent academic performance measure). Given $\mathbf{x}_i$, SHARE produces a primary classification output $\hat{y}_i^{\text{cls}}$ and an auxiliary GPA estimate $\hat{y}_i^{\text{gpa}}$. Throughout the paper, classification is the main task of interest because it offers a thresholded view of academic or mental-health risk, while the GPA target is retained only as secondary supervision and as a continuous diagnostic of academic performance.

Algorithm 1: Leakage-audited training pipeline of SHARE

Input : Audited features $\mathbf{x}_i = (\{\mathbf{x}_i^{(m)}\}_{m\in\mathcal{M}_i}, \mathbf{x}_i^u, \mathbf{x}_i^d)$, labels $y_i^{\text{cls}}, y_i^{\text{gpa}}$ for $i = 1, \ldots, N$; hyperparameters d, H, α.
Output: Predictions $\hat{\mathbf{y}}_{\text{cls}}, \hat{\mathbf{y}}_{\text{gpa}}$; joint loss $\mathcal{L}_{\text{total}}$.

```
// Stage 1: Mental Health Feature Extractor (MHE)
for i ← 1 to N do
```

$\mathbf{t}_i^c \leftarrow \frac{1}{|\mathcal{M}_i|} \sum_{m\in\mathcal{M}_i} \texttt{LayerNorm}(\mathbf{W}_m \mathbf{x}_i^{(m)} + \mathbf{b}_m)$; // Clinical token, Eq. 3-4

$\mathbf{t}_i^u \leftarrow \texttt{LayerNorm}(\mathbf{W}_u \mathbf{x}_i^u + \mathbf{b}_u)$; $\mathbf{t}_i^d \leftarrow \texttt{LayerNorm}(\mathbf{W}_d \mathbf{x}_i^d + \mathbf{b}_d)$; // Eq. 5-6

$\mathbf{T}_i \leftarrow [\mathbf{t}_i^c; \mathbf{t}_i^u; \mathbf{t}_i^d]$; // Token matrix, Eq. 7

```
// Stage 2: Service Utilization Attention (SUA)
for h ← 1 to H do
```

$\mathbf{q}_{i,h}, \mathbf{K}_{i,h}, \mathbf{V}_{i,h} \leftarrow \mathbf{t}_i^u \mathbf{W}_h^Q,\ \mathbf{T}_i \mathbf{W}_h^K,\ \mathbf{T}_i \mathbf{W}_h^V$; // Eq. 8-10

$\mathbf{u}_{i,h} \leftarrow \texttt{Softmax}(\mathbf{q}_{i,h} \mathbf{K}_{i,h}^\top / \sqrt{d_h})\, \mathbf{V}_{i,h}$; // Eq. 11-12

```
end
```

$\tilde{\mathbf{t}}_i^u \leftarrow \texttt{LayerNorm}(\mathbf{t}_i^u + [\mathbf{u}_{i,1}; \ldots; \mathbf{u}_{i,H}]\mathbf{W}^O)$; // Eq. 13

$\mathbf{h}_i \leftarrow \texttt{LayerNorm}(\mathbf{W}_f [\mathbf{t}_i^c; \tilde{\mathbf{t}}_i^u; \mathbf{t}_i^d] + \mathbf{b}_f)$; // Fused embedding, Eq. 14

```
// Stage 3: Dual-task Academic Predictor (DAP)
```

$\hat{y}_i^{\text{cls}} \leftarrow \texttt{Sigmoid}(\mathbf{W}_{\text{cls}}\, \mathbf{h}_i + b_{\text{cls}})$; $\hat{y}_i^{\text{gpa}} \leftarrow \mathbf{W}_{\text{gpa}}\, \mathbf{h}_i + b_{\text{gpa}}$; // Eq. 15-16

```
end
// Stage 4: Classification-first joint objective
```

$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{cls}}(\hat{\mathbf{y}}_{\text{cls}}, \mathbf{y}_{\text{cls}}) + \alpha\, \mathcal{L}_{\text{reg}}(\hat{\mathbf{y}}_{\text{gpa}}, \mathbf{y}_{\text{gpa}})$; // Eq. 17-19

return $\hat{\mathbf{y}}_{\text{cls}},\ \hat{\mathbf{y}}_{\text{gpa}},\ \mathcal{L}_{\text{total}}$

Training therefore optimizes a classification-first objective with an auxiliary regression term:

$$\theta^* = \arg\min_{\theta}\ \mathbb{E}_{(\mathbf{x}, y^c, y^g)\sim\mathcal{D}}[\mathcal{L}_{\text{cls}}(f_\theta(\mathbf{x}), y^c) + \alpha\, \mathcal{L}_{\text{reg}}(f_\theta(\mathbf{x}), y^g)], \tag{2}$$

where $\alpha \geq 0$ controls the contribution of the auxiliary regression loss. This weighting is not meant to elevate GPA prediction to a co-primary empirical claim; rather, it allows the shared encoder to receive additional academic-performance supervision while evaluation remains centered on the binary risk patterns used to probe support-service association.

Crucially, the definition of y_i^{cls} is dataset-specific because the public benchmarks require leakage-aware reformulation. On MHP-Multi, y_i^{cls} denotes PHQ/GAD-based mental-health risk, but all PHQ-9 and GAD-7 items and totals are excluded from $\mathbf{x}_i$, so prediction must rely on the remaining PSS-10, demographic, and support-related covariates. On SMH-Kaggle, y_i^{cls} denotes academic risk (CGPA < 2.5), and CGPA is excluded from $\mathbf{x}_i$, leaving only symptom indicators, treatment-seeking variables, and demographics as predictors. These leak-audited task definitions are part of the problem specification itself rather than a downstream implementation detail, because they enforce separation between labels and inputs before any architectural comparison is made.

3.3. Mental Health Feature Extractor (MHE)

Clinical scale scores from different instruments occupy different numeric ranges and encode distinct latent constructs. PHQ-9 scores span $[0, 27]$, GAD-7 $[0, 21]$, PSS-10 $[0, 40]$, and BDI-II $[0, 63]$—a four-fold variation in scale that can distort downstream similarity computations. MHE resolves this by converting each student record into three fixed-width semantic tokens: a clinical token, a service-utilization token, and a context token.

For student i, we partition the raw input into instrument-specific clinical subsets $\{\mathbf{x}_i^{(m)}\}_{m\in\mathcal{M}_i}$, service features $\mathbf{x}_i^u$, and contextual covariates $\mathbf{x}_i^d$. Each available clinical instrument is projected independently:

$$\mathbf{e}_i^{(m)} = \text{LayerNorm}\Big(\mathbf{W}_m \mathbf{x}_i^{(m)} + \mathbf{b}_m\Big), \quad \mathbf{e}_i^{(m)} \in \mathbb{R}^d, \tag{3}$$

where $\mathcal{M}_i$ denotes the set of clinical instruments observed for student i. The clinical token is then computed as the mean of available instrument embeddings:

$$\mathbf{t}_i^c = \frac{1}{|\mathcal{M}_i|} \sum_{m \in \mathcal{M}_i} \mathbf{e}_i^{(m)}, \quad \mathbf{t}_i^c \in \mathbb{R}^d. \tag{4}$$

Service-utilization and contextual features are encoded separately:

$$\mathbf{t}_i^u = \text{LayerNorm}(\mathbf{W}_u \mathbf{x}_i^u + \mathbf{b}_u), \tag{5}$$

$$\mathbf{t}_i^d = \text{LayerNorm}\Big(\mathbf{W}_d \mathbf{x}_i^d + \mathbf{b}_d\Big), \tag{6}$$

with $\mathbf{t}_i^u, \mathbf{t}_i^d \in \mathbb{R}^d$. The resulting per-student token matrix is

$$\mathbf{T}_i = [\mathbf{t}_i^c;\ \mathbf{t}_i^u;\ \mathbf{t}_i^d] \in \mathbb{R}^{3 \times d}. \tag{7}$$

This factorization preserves instrument-specific signal while preventing large-scale instruments from dominating the downstream interaction mechanism. It also makes the representation consistent across datasets: if a benchmark lacks a fine-grained service category, the corresponding information is folded into $\mathbf{x}_i^u$ as a coarse utilization indicator rather than forcing an incompatible feature schema.

Handling heterogeneous feature types. Our experiments encode clinical information in two different forms. MHP-Multi contributes ten PSS-10 item responses, which are treated as a continuous clinical block after standardization. SMH-Kaggle contributes three binary symptom indicators (depression, anxiety, panic attack), which are projected directly without normalization because they already lie in [0, 1]. The LayerNorm applied after each projection (Eq. 3) equalizes output magnitude regardless of input scale, ensuring that neither binary nor continuous inputs dominate the clinical token. If a student record provides no clinical instrument measurements ($|\mathcal{M}_i| = 0$), the clinical token defaults to a learned, zero-initialized embedding $\mathbf{t}_i^c = \mathbf{0}_d$; this degenerate case does not arise in the benchmarks studied here.

3.4. Service Utilization Attention (SUA)

The distinctive inductive bias in SHARE is SUA, which treats service utilization as the anchor signal rather than as one feature group among many. Instead of letting all semantic tokens interact symmetrically, SUA uses the service token to query the clinical and contextual tokens for evidence about how a student's current support pattern aligns with mental-health burden and academic risk.

Why the service token as anchor? In the mental health services domain, service utilization is categorically different from clinical symptoms or demographic covariates: it is an institutional response variable rather than only a descriptor of student state. Existing tabular Transformers—including FT-Transformer and SAINT—apply symmetric self-attention that treats all feature groups with equal agency, allowing clinical score tokens to query service tokens just as readily as the reverse. This symmetry is appropriate for general-purpose prediction but less aligned with the descriptive question we are answering: "given this student's service engagement pattern, how does it align with their clinical burden and academic-risk profile?" Anchoring the query to the service token encodes this directionality explicitly. The resulting attention weights $\mathbf{a}_{i,h} \in \mathbb{R}^{1 \times 3}$ are semantically interpretable: they represent how much each head weights clinical burden, service utilization, and contextual evidence *from the service perspective*—a compact operationalization of service-outcome association analysis.

Distinction from generic CLS-token designs. Superficially, using a fixed token as the query resembles the CLS token in BERT-style encoders. However, the CLS token is a task-agnostic aggregator trained to summarize all input content without domain preference, and it queries all other tokens symmetrically via full bidirectional attention. SUA differs in two respects. First, the query token is domain-selected rather than task-agnostic: it corresponds specifically to service utilization features, whose embedding encodes the modality and dosage of institutional support. Second, the key-value context includes all three tokens—clinical, service, and contextual—so the service token can attend back to its own embedding, allowing SUA to model self-consistency in the service representation (e.g., whether high utilization aligns with or diverges from clinical severity). This design provides inductive bias without removing flexibility: if the service signal is uninformative, the attention weights simply converge toward uniform, reducing SUA to standard averaging.

For each student i and attention head h, we compute

$$\mathbf{q}_{i,h} = \mathbf{t}_i^u \mathbf{W}_h^Q, \tag{8}$$

$$\mathbf{K}_{i,h} = \mathbf{T}_i \mathbf{W}_h^K, \tag{9}$$

$$\mathbf{V}_{i,h} = \mathbf{T}_i \mathbf{W}_h^V, \tag{10}$$

where $\mathbf{q}_{i,h} \in \mathbb{R}^{1\times d_h}$, $\mathbf{K}_{i,h}, \mathbf{V}_{i,h} \in \mathbb{R}^{3\times d_h}$, and $d_h = d/H$. The attention weights are

$$\mathbf{a}_{i,h} = \mathrm{softmax}\left(\frac{\mathbf{q}_{i,h}\mathbf{K}_{i,h}^\top}{\sqrt{d_h}}\right) \in \mathbb{R}^{1\times 3}, \tag{11}$$

yielding the head-specific update

$$\mathbf{u}_{i,h} = \mathbf{a}_{i,h}\mathbf{V}_{i,h}. \tag{12}$$

The updated service token is obtained by residual fusion across heads:

$$\tilde{\mathbf{t}}_i^u = \mathrm{LayerNorm}\Big(\mathbf{t}_i^u + [\mathbf{u}_{i,1}; \ldots; \mathbf{u}_{i,H}]\mathbf{W}^O\Big), \tag{13}$$

and the final student representation is

$$\mathbf{h}_i = \mathrm{LayerNorm}\Big(\mathbf{W}_f[\mathbf{t}_i^c;\, \tilde{\mathbf{t}}_i^u;\, \mathbf{t}_i^d] + \mathbf{b}_f\Big), \tag{14}$$

where $\mathbf{W}^O \in \mathbb{R}^{d\times d}$ and $\mathbf{W}_f \in \mathbb{R}^{d\times 3d}$. This design yields a compact interpretable attention pattern: each head produces three weights describing how much clinical burden, service uptake, and context contribute to the refined service representation. In other words, SUA does not simply ask "which feature is important?"; it asks "given the current service pattern, which surrounding evidence should govern the academic-risk estimate?"

3.5. Dual-task Academic Predictor (DAP)

DAP consists of two linear prediction heads applied to the shared representation $\mathbf{H}$. The classification head produces an at-risk score:

$$\hat{y}_i^{\mathrm{cls}} = \mathrm{sigmoid}(\mathbf{W}_{\mathrm{cls}}\,\mathbf{h}_i + b_{\mathrm{cls}}), \tag{15}$$

where $\mathbf{h}_i \in \mathbb{R}^d$ is the attended representation for student i, $\mathbf{W}_{\mathrm{cls}} \in \mathbb{R}^{1\times d}$, and $b_{\mathrm{cls}} \in \mathbb{R}$. The regression head predicts GPA:

$$\hat{y}_i^{\mathrm{gpa}} = \mathbf{W}_{\mathrm{gpa}}\,\mathbf{h}_i + b_{\mathrm{gpa}}, \tag{16}$$

where $\mathbf{W}_{\mathrm{gpa}} \in \mathbb{R}^{1\times d}$ and $b_{\mathrm{gpa}} \in \mathbb{R}$.

The two heads share the MHE and SUA encoder, which requires the learned representation to support both at-risk detection and GPA estimation. In principle, this sharing can act as a form of multi-task regularization by exposing the encoder to both thresholded and continuous supervision, although in the present benchmarks the GPA signal remains weak and this benefit is limited in practice.

3.6. Training Objective

The joint training objective combines binary cross-entropy for classification and mean absolute error for regression:

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{N}\sum_{i=1}^{N}[y_i^c \log \hat{y}_i^c + (1 - y_i^c)\log(1 - \hat{y}_i^c)], \tag{17}$$

$$\mathcal{L}_{\mathrm{reg}} = \frac{1}{N}\sum_{i=1}^{N}\left|\hat{y}_i^g - y_i^g\right|, \tag{18}$$

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{cls}} + \alpha\,\mathcal{L}_{\mathrm{reg}}, \tag{19}$$

where $\alpha = 0.5$ balances classification and regression contributions in the default configuration. We use the Adam optimizer (Kingma and Ba, 2015) with learning rate 1×10^{-3}, batch size 32, and train for 100 epochs with early stopping (patience = 20) on validation loss. Class imbalance in the classification target is addressed via class-weighted cross-entropy, with weights inversely proportional to class frequency. Throughout the empirical study, classification is the primary evaluation focus and regression is reported as a secondary diagnostic.

Table 1: Leakage-audited tasks used to study support-service and academic-risk associations.

Dataset	Task	N	Pos.%	CV
SMH-Kaggle	Academic risk (CGPA < 2.5)	101	5.9%	3-fold
MHP-Multi	PHQ/GAD risk from non-PHQ/GAD covariates	2,028	79.0%	5-fold

3.7. Ethics and Data Governance

This study analyzes two publicly available, de-identified datasets and does not involve new participant recruitment, intervention, or access to identifiable student records by the authors. The work is therefore reported as a secondary analysis of public data rather than as a new human-subject experiment. For the same reason, the results are interpreted as descriptive and correlational rather than as evidence of counseling effectiveness or causal impact.

3.8. Leakage-Aware Tasks and Evaluation Protocol

We evaluate SHARE on two public datasets, but we do not reuse their original task formulations uncritically. Every reported classification result is obtained under an explicit leakage audit so that the remaining estimates can be interpreted as evidence about support-service and outcome association rather than direct label reconstruction.

SMH-Kaggle: The raw dataset contains 101 students from a Malaysian university with binary symptom indicators (depression, anxiety, panic attack), treatment-seeking behavior, demographic variables, and CGPA. Instead of predicting a label partly defined by the CGPA feature itself, we define the classification task as *academic-risk prediction*: positive if CGPA < 2.5, while CGPA is excluded from the model input. This leaves only six positive examples, so the dataset becomes a severe low-prevalence benchmark.

MHP-Multi: The raw dataset contains 2,028 students from 15 Bangladeshi universities with PHQ-9, GAD-7, PSS-10, demographic variables, and institutional-support indicators. The original at-risk label is defined by PHQ-9 or GAD-7 thresholds. To remove leakage, we retain this label for evaluation but exclude all PHQ-9 and GAD-7 totals and item responses from the input. The resulting input consists only of PSS-10 item responses, demographics, and a scholarship/support proxy.

Table 1 summarizes the resulting tasks. Section 4 then evaluates model behavior under these audited definitions.

Evaluation protocol. We report mean F_1 as the primary classification metric, with AUC as a threshold-independent companion metric. Tables report mean ± standard deviation across fold-level results after averaging each fold over three random seeds (1, 42, 2025). SMH-Kaggle uses 3-fold stratified cross-validation because the academic-risk task contains only six positives; MHP-Multi uses 5-fold stratified cross-validation. GPA regression is retained as a secondary analysis and evaluated with MAE and R^2.

3.9. Baselines

We compare against 12 baselines spanning three families. **Classical models:** LR, SVM, and NB. **Tree/ML baselines:** Random Forest, LightGBM, XGBoost, CatBoost, and MLP. **Tabular deep learning baselines:** TabTransformer, FT-Transformer, SAINT, and TabPFN v2 (Hollmann et al., 2024) (a meta-trained in-context learning model; we include it because the background review discusses it and both datasets satisfy the $N \leq 10{,}000$ requirement). These baselines serve as robustness checks for the substantive interpretation of the data rather than only as model-ranking competitors. For the regression table we additionally report Ridge and Random-Forest regression baselines, since the classification-only baselines do not expose a comparable GPA head.

4. Results

4.1. Leakage-Audited Relationship Analysis and Classification Results

The first result is that apparent benchmark strength drops sharply once direct label-feature overlap is removed. We therefore center the analysis on the leak-free task definitions described in Section 3.8 and report the audited classification results in Table 2.

Table 2: Leakage-audited classification results. Values are mean ± standard deviation across stratified CV folds after seed averaging. **Bold** marks the best mean value in each column; underlined marks the second-best mean F_1 in each dataset. The large standard deviations on SMH-Kaggle reflect the instability of a 6-positive task.

Model	SMH-Kaggle (Academic-risk)		MHP-Multi (PHQ/GAD-risk)	
	F_1	AUC	F_1	AUC
LR	0.061 ± 0.086	0.688 ± 0.099	0.807 ± 0.020	0.829 ± 0.022
SVM	0.000 ± 0.000	0.472 ± 0.047	0.888 ± 0.007	0.809 ± 0.024
NB	0.216 ± 0.019	**0.740 ± 0.089**	0.846 ± 0.017	0.829 ± 0.015
Random Forest	0.133 ± 0.189	0.667 ± 0.136	0.883 ± 0.008	0.812 ± 0.020
MLP	0.000 ± 0.000	0.527 ± 0.179	0.891 ± 0.005	0.824 ± 0.020
LightGBM	0.115 ± 0.003	0.465 ± 0.060	0.833 ± 0.018	0.815 ± 0.020
XGBoost	0.000 ± 0.000	0.566 ± 0.042	0.892 ± 0.007	0.822 ± 0.018
CatBoost	**0.241 ± 0.205**	0.654 ± 0.160	0.829 ± 0.017	0.833 ± 0.016
TabTransformer	0.095 ± 0.135	0.730 ± 0.066	0.847 ± 0.010	0.786 ± 0.026
FT-Transformer	0.150 ± 0.108	0.707 ± 0.110	0.823 ± 0.014	0.800 ± 0.012
SAINT	0.095 ± 0.135	0.698 ± 0.085	0.822 ± 0.002	0.779 ± 0.038
TabPFN v2	0.000 ± 0.000	0.645 ± 0.137	**0.902 ± 0.006**	**0.836 ± 0.017**
SHARE (ours)	0.132 ± 0.186	0.668 ± 0.053	0.825 ± 0.017	0.797 ± 0.020

Fold-matched diagnostic. Using one seed-averaged SHARE F_1 score per fold and a matched baseline score per fold, two-sided Wilcoxon signed-rank tests yield $p = 0.6547$ for SHARE vs. CatBoost on SMH-Kaggle ($n = 3$ folds) and $p = 0.0625$ for SHARE vs. TabPFN v2 on MHP-Multi ($n = 5$ folds). These low-power checks are reported only as descriptive fold-matched diagnostics, not as confirmatory significance evidence, because the audited benchmarks provide only 3 and 5 paired folds respectively.

Figure 2 summarizes the leak-free MHP-Multi leaderboard, where the service-aware analytical framework sits in the middle of the pack rather than defining a new frontier.

MHP-Multi. The most important result is that near-perfect performance does not survive leakage auditing. Once PHQ/GAD-derived inputs are removed, the strongest audited models reach about $F_1 = 0.90$, and strong generic baselines remain ahead of SHARE. After adding a modern small-data tabular baseline, the strongest models are TabPFN v2 ($F_1 = 0.902$, AUC = 0.836), XGBoost (0.892), and MLP (0.891). SHARE reaches $F_1 = 0.825$, comparable to CatBoost (0.829), FT-Transformer (0.823), and SAINT (0.822), but still clearly below the strongest baselines. The fold-matched Wilcoxon diagnostic is directionally consistent with the leaderboard ($p = 0.0625$), but with only five paired folds it should be read as descriptive rather than confirmatory. Substantively, this means that the remaining stress, demographic, and support-access variables retain only moderate association with PHQ/GAD-defined risk once leakage is removed, and the current service-aware inductive bias does not reveal stronger incremental service signal than well-performing generic tabular models already capture.

SMH-Kaggle. The academic-risk task is extremely small and imbalanced: only 6 of 101 students satisfy CGPA < 2.5. Under this setting, all methods are unstable and absolute F_1 values are low. CatBoost achieves the best mean $F_1 = 0.241 \pm 0.205$, while NB attains perfect recall with the best mean AUC (0.740 ± 0.089) at the cost of low precision. Several methods, including SVM, MLP, XGBoost, and TabPFN v2, collapse to predicting only the majority class. SHARE reaches $F_1 = 0.132 \pm 0.186$ with AUC = 0.668 ± 0.053, which remains below the strongest baselines. The corresponding fold-matched Wilcoxon diagnostic ($p = 0.6547$) is likewise only descriptive, because the audited task yields just three paired folds. In a 6-positive setting, small preprocessing or fold differences can shift the ranking, which further reinforces that SMH-Kaggle behaves as a high-variance stress test rather than a stable basis for inferring how mental health support variables relate to academic risk.

4.2. Hyperparameter Sensitivity Results

To test whether the substantive interpretation depends on one unusually favorable SHARE configuration, we ran one-factor sensitivity sweeps on leak-free MHP-Multi while holding all remaining settings fixed at the paper's default configuration ($d = 128$, $H = 8$, $\alpha = 0.5$). To keep this follow-up lightweight, the sweep uses a reduced diagnostic protocol with one seed per fold rather than the full 3-seed averaging used in Table 2; the goal is to assess trend consistency rather than exact numerical parity with the main leaderboard. We vary the auxiliary-loss weight α, the number of attention heads, and the representation width d, and report the same F_1, AUC, and MAE metrics used elsewhere in the paper. Table 3 summarizes the sweep values, while Figure 3 visualizes the corresponding trade-offs.

The sweep over α mainly tests whether the GPA branch contributes useful auxiliary supervision or simply acts as an unused extra head. Moving from $\alpha = 0$ to $\alpha = 0.5$ improves F_1 from 0.807 to 0.822, while the MAE collapse from

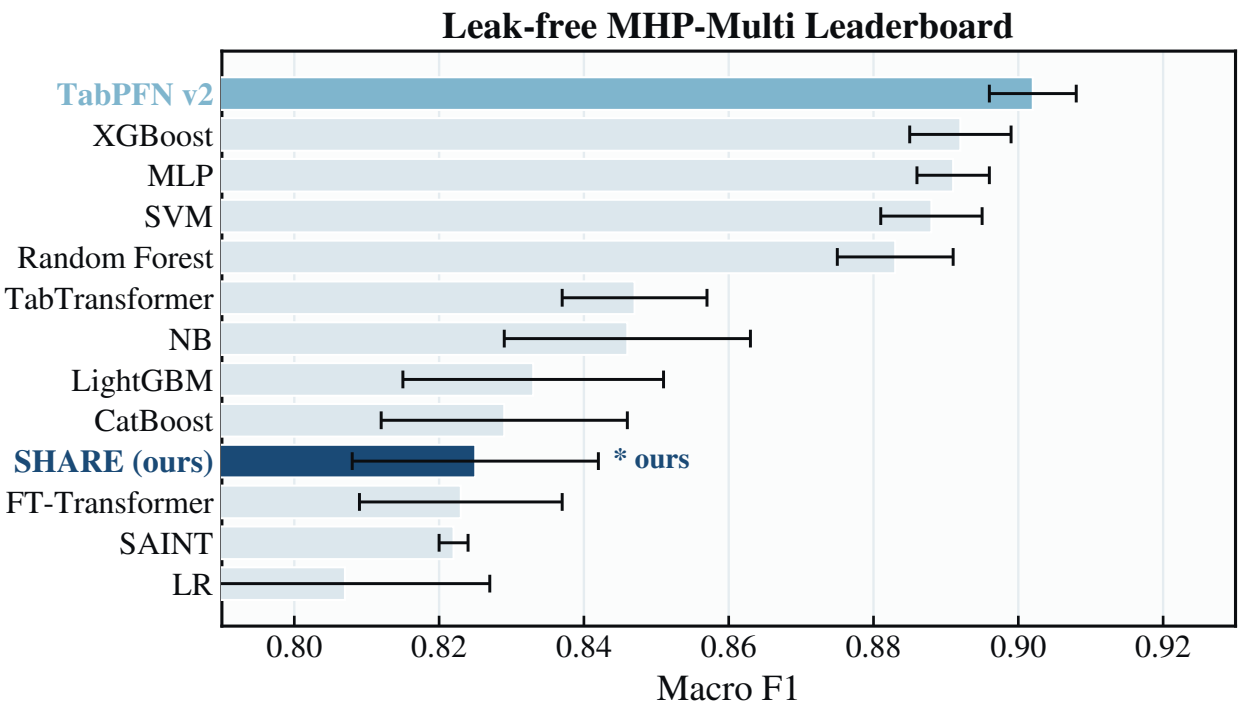

Figure 2: Leak-free macro-F_1 on MHP-Multi across all evaluated models (mean ± s.d. across seed-averaged CV folds). SHARE is highlighted in dark navy and the strongest baseline (TabPFN v2) in light blue.

3.303 to 0.298 is expected because the regression head is unsupervised when $\alpha = 0$. Increasing to $\alpha = 1.0$ yields only a marginal further gain in F_1 and MAE while leaving AUC nearly unchanged, so we keep $\alpha = 0.5$ as a conservative default rather than retuning to maximize a single audited benchmark.

Changing the number of attention heads produces a different pattern: the overall variation remains modest, but the preferred operating point depends on which metric is emphasized. The default 8-head setting attains the highest F_1 among the tested values, whereas a single head gives the strongest AUC. This divergence suggests that richer interaction mainly changes the thresholded decision profile of the model, rather than uniformly improving global score ranking.

Representation width is similarly non-monotonic. Small-to-intermediate widths (64–128) form a narrow plateau for F_1 and MAE, whereas larger widths (256–512) raise AUC slightly but depress F_1 and worsen MAE. On this small audited task, additional capacity therefore appears to increase ranking flexibility more than it improves the calibrated decision boundary. We retain $d = 128$ for consistency with the cross-dataset default used throughout the paper; on leak-free MHP-Multi, 64 and 128 behave as a similar performance plateau under this reduced sweep.

4.3. Ablation Results

We next isolate which parts of SHARE materially affect the audited MHP-Multi task. Table 4 compares the full model with three controlled variants: replacing SUA with uniform token fusion (*w/o SUA*), disabling auxiliary GPA supervision by setting $\alpha = 0$, and replacing the hierarchical MHE encoder with the raw-group encoder used in the simplest variant of the architecture. Because the $\alpha = 0$ setting removes the regression loss rather than the prediction head itself, its MAE should be read only as a diagnostic consequence of leaving the GPA output unsupervised. These ablations are intended to test robustness of the relationship analysis rather than to claim component-wise superiority in isolation.

The full model attains the best cross-validated F_1 among its own variants, but the degradation pattern is uneven rather than uniformly favorable to every proposed component. Replacing SUA yields the largest F_1 drop, from 0.825 to 0.804, which indicates that service-aware fusion improves the default thresholded operating point. At the same

Table 3: One-factor hyperparameter sensitivity for SHARE on leak-free MHP-Multi under a reduced diagnostic protocol. Boldface marks the default configuration used in this sweep.

Sweep	Setting	F_1	AUC	MAE
α	0.0	0.807	0.799	3.303
	0.5	0.822	0.793	0.298
	1.0	0.824	0.800	0.297
n_{heads}	1	0.818	0.827	0.298
	4	0.807	0.805	0.297
	8	0.822	0.793	0.298
d	64	0.823	0.798	0.292
	128	0.820	0.800	0.292
	256	0.813	0.815	0.298
	512	0.790	0.818	0.304

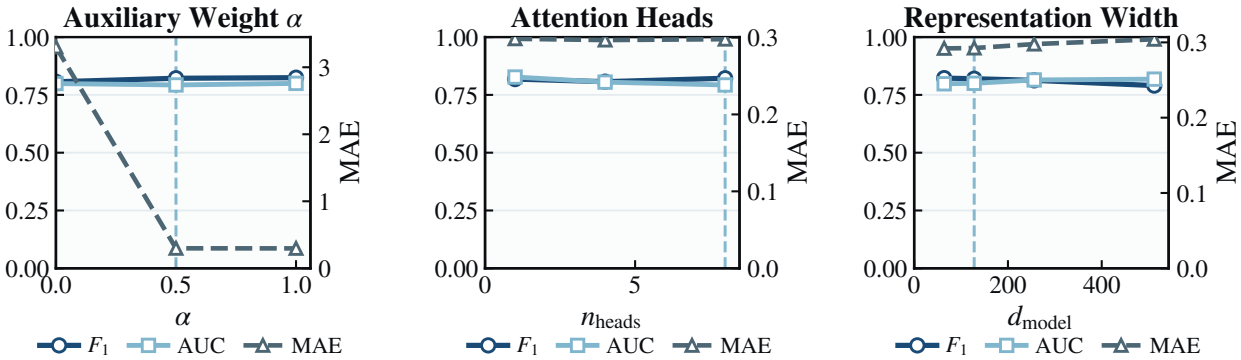

Figure 3: Hyperparameter sensitivity on leak-free MHP-Multi under a reduced diagnostic protocol. Each panel varies a single hyperparameter while keeping the other settings fixed at the default configuration; solid lines use the left axis (F_1 and AUC) and the dashed line uses the right axis (MAE). The vertical dashed line marks the default configuration used in this sweep rather than an exact numerical duplicate of the main 3-seed leaderboard.

time, the same variant attains the highest AUC and slightly lower MAE, so SUA cannot be interpreted as a universal gain in class separability or auxiliary regression. Its effect is more specific: it shifts how the model trades off positives and negatives under the fixed 0.5 threshold used throughout evaluation.

Disabling auxiliary GPA supervision by setting $\alpha = 0$ lowers F_1 to 0.812 while leaving AUC close to the full model, which suggests that the secondary target provides a useful but limited regularizing signal for classification. The accompanying MAE increase to 3.303 is expected once the regression output is no longer supervised, and should therefore not be read as a comparable regression result in the same sense as the other rows. Replacing MHE with the raw-group encoder yields a similar F_1 level (0.813) but the weakest AUC (0.790), indicating that hierarchical group encoding helps preserve a more coherent representation across the clinical, service, and contextual inputs, albeit with a moderate rather than dominant effect size; Figure 4 provides a complementary view of this pattern on one representative split and seed.

4.4. Oracle Feature-Set Results on MHP-Multi

A natural question is whether the limited ceiling on MHP-Multi reflects architectural constraints or simply the absence of support-service signal beyond what PSS-10 alone can provide. To answer this, we ran an incremental feature-set analysis using XGBoost (a strong tree-based baseline and the best conventional model before adding TabPFN v2) under the same 5-fold CV × 3-seed protocol, and Table 5 reports the resulting three-tier comparison.

Table 5 shows that PSS-10 stress items alone already reach the performance ceiling of the full 14-feature task ($F_1 = 0.892$). Adding demographic covariates and the service proxy produces no measurable gain in either F_1 or

Table 4: Ablation-style diagnostics for SHARE on leak-free MHP-Multi. ΔF_1 is measured relative to the full model.

Variant	F_1	ΔF_1	AUC	MAE
SHARE (full)	**0.825**	0.000	0.797	0.299
w/o SUA	0.804	−0.021	**0.832**	**0.294**
$\alpha = 0$	0.812	−0.013	0.803	3.303
w/o MHE	0.813	−0.012	0.790	0.299

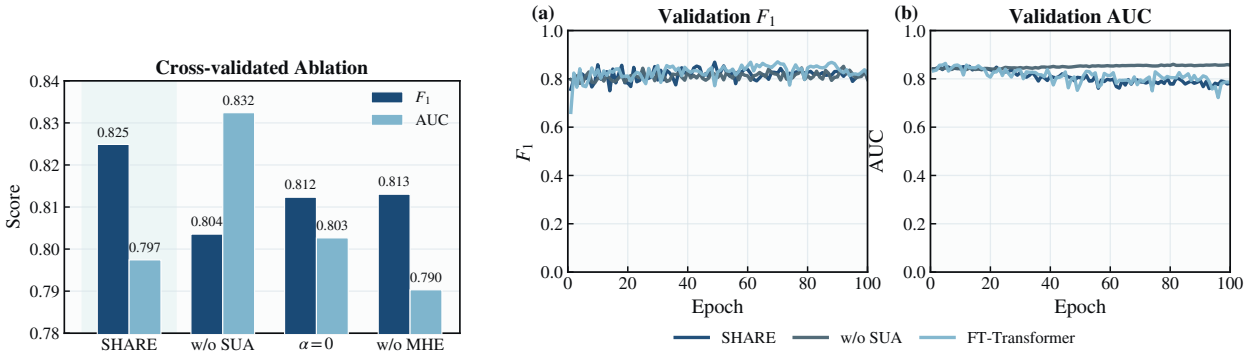

Figure 4: Follow-up diagnostics on leak-free MHP-Multi. Left: cross-validated F_1 and AUC for the full model and three controlled variants. Right: validation F_1 and AUC across epochs on one representative split and one seed for SHARE, *w/o SUA*, and FT-Transformer; this panel is included as an illustrative training-behavior diagnostic rather than a cross-validated stability estimate. The no-SUA variant favors AUC, whereas the full model retains the strongest cross-validated F_1 within its own family.

AUC. This suggests that the current public MHP-Multi service variable contributes little incremental association once stress burden is observed, which substantially limits how strongly the dataset can support claims about support-service relevance to student outcomes.

4.5. Brief GPA Regression Check

Table 6 reports GPA regression results using the same leak-free inputs. We include this only as a brief secondary check because the public datasets provide limited signal for accurate GPA estimation from the remaining covariates.

The regression results remain modest across both datasets, with simple baselines outperforming SHARE on the available leak-free features. This brief check therefore does not alter the main leakage-audit conclusion. In substantive terms, current public support-service and symptom covariates do not support strong continuous inference about academic performance once leakage-prone variables are removed.

5. Discussion

5.1. Principal Findings

The central question of this study is whether current public higher-education datasets can support credible analysis of the relationship between mental health support services and academic performance. After the leakage audit summarized in Appendix Table 7, the answer is only partly. On the leak-free MHP-Multi task, the remaining stress, demographic, and support-access variables retain moderate association with PHQ/GAD-defined risk, but strong generic baselines remain ahead. On the SMH-Kaggle academic-risk task, all methods remain unstable because only six positive examples survive the audit. GPA regression is also weak, and the oracle feature-set analysis on MHP-Multi shows that the current service proxy adds essentially no measurable gain beyond PSS-10 items alone.

These results shift the paper's contribution away from architectural competition and toward a more cautious substantive interpretation. Current public higher-education datasets provide only limited correlational evidence about how mental health support services relate to academic performance or academic risk. Once direct label proxies are removed, the available service indicators are either too coarse (MHP-Multi) or embedded in too few positive cases

Table 5: Oracle feature-set analysis on MHP-Multi. XGBoost is evaluated with three nested feature tiers. Values are mean ± standard deviation.

Tier	Feature set	#Feat.	F_1	AUC
Tier 1	PSS-10 items only	10	0.892 ± 0.009	0.824 ± 0.020
Tier 2	PSS-10 + demographics	13	0.893 ± 0.010	0.823 ± 0.021
Tier 3	PSS-10 + demographics + service	14	0.893 ± 0.010	0.823 ± 0.024

Table 6: Brief GPA regression check (mean ± standard deviation; lower MAE and higher R^2 are better).

Model	SMH-Kaggle		MHP-Multi	
	MAE	R^2	MAE	R^2
Ridge	**0.405 ± 0.028**	**−0.178 ± 0.168**	**0.285 ± 0.008**	**0.066 ± 0.029**
Random Forest Regr.	0.423 ± 0.026	−0.142 ± 0.270	0.287 ± 0.006	0.018 ± 0.038
SHARE (ours)	0.492 ± 0.099	−0.889 ± 0.997	0.299 ± 0.007	−0.061 ± 0.043

(SMH-Kaggle) to support strong claims. We therefore argue that benchmark audit protocols—systematically identifying and removing label-feature overlap, and reporting the resulting performance drop before comparing models—should be a standard reporting requirement in educational data mining and health informatics publications.

That said, SHARE still provides a useful analytical template. Its semantic grouping of clinical, service, and contextual inputs remains well motivated for student-support data, and the service-anchored attention mechanism offers a structured way to encode service-related variables rather than treating them as ordinary covariates. Our follow-up diagnostics show that the full SHARE configuration attains the best cross-validated F_1 among its own ablations on MHP-Multi, but the gain over neighboring variants is modest and does not close the gap to strong generic baselines. Notably, removing SUA lowers F_1 while raising AUC, suggesting that the module changes threshold behavior more than ranking ability. In the present data regime, the inductive bias is therefore interpretable but not performance defining.

5.2. Ethical Considerations and Fairness

Analytical models for student mental health and academic risk carry significant ethical responsibilities. Three issues remain central.

Selection bias and causal interpretation. SHARE is trained on observational cross-sectional data. Students who access support services may differ systematically from those who do not, so model outputs should be interpreted as correlational risk indicators rather than causal estimates of service effectiveness or academic improvement. Institutions should not use these patterns as evidence that a specific intervention causes better outcomes without additional causal analysis.

Subgroup fairness. Aggregate performance can hide systematic disparities across gender, race/ethnicity, first-generation status, or institution type. Appendix Table 8 reports a preliminary gender-sliced diagnostic on leak-free MHP-Multi. In that descriptive comparison, both SHARE and TabPFN v2 perform worse on the male subgroup than on the female subgroup, and SHARE shows materially higher false-negative rates than the baseline on both major gender slices. Before any deployment, subgroup-stratified F_1, false-negative rate, and calibration error should be reported on the target population. This is particularly important in student-support settings, where false negatives correspond to vulnerable students who are missed by outreach workflows. Because this appendix diagnostic comes from one audited dataset and the "Prefer not to say" slice contains only 10 students, it should be treated as preliminary rather than definitive.

Human oversight. The intended use of SHARE is decision support, not autonomous action. Any deployment should keep counselors or advisors in the loop, document the limited evidentiary scope of the predictions, and restrict access to appropriately governed student data.

5.3. Limitations

Several limitations remain. **Benchmark scope:** the paper evaluates only two public datasets, both cross-sectional and relatively small. **Task fragility:** SMH-Kaggle becomes a rare-event problem after CGPA leakage is removed,

while MHP-Multi still uses a coarse support-access proxy rather than detailed service histories. **Regression weakness:** the GPA signal available in the remaining features is limited, so the dual-task formulation is not empirically justified by the current public data alone. **Framework competitiveness:** the experiments show that generic baselines remain stronger than SHARE on both datasets, which means the framework should currently be understood as exploratory rather than performance defining.

5.4. Future Directions

The clearest path forward is richer data. Future work should evaluate service-aware models on longitudinal cohorts with richer support logs: session counts, modality sequences, referral source, and timing relative to academic milestones. A second priority is larger leak-free student outcome datasets with enough positive cases to support stable thresholded evaluation. Third, future studies should report calibration, subgroup fairness, and uncertainty alongside discrimination metrics. With those ingredients in place, service-aware architectures such as SHARE can be tested in a setting where their inductive bias has a realistic chance to matter.

CRediT authorship contribution statement

Mira Chen: Conceptualization, Methodology, Formal analysis, Software, Writing—original draft, Visualization.
Ethan Zhao: Data curation, Investigation, Writing—review & editing.
Lena Park: Supervision, Project administration, Validation, Writing—review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets analyzed in this study are publicly available. The **Student Mental Health Dataset** (SMH-Kaggle) is available via Kaggle: https://www.kaggle.com/datasets/shariful07/student-mental-health. The **Comprehensive Mental Health Problems Dataset** (MHP-Multi) is available via figshare: https://figshare.com/articles/dataset/MHP_Anxiety_Stress_Depression_Dataset_of_University_Students/25771164/1. Analysis code supporting the findings is available from the corresponding author upon reasonable request.

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Appendix

Leakage audit details. Table 7 lists the label definitions and the inputs removed during the audit. These design choices are the main methodological difference between the leakage-audited protocol used here and the more permissive benchmark formulations often adopted in prior work.
Implementation details. SHARE is implemented in PyTorch (Paszke et al., 2019). We use $d = 128$, $H = 8$, dropout $= 0.2$, $\alpha = 0.5$, batch size = 32, learning rate $= 1 \times 10^{-3}$, maximum epochs = 100, and early stopping patience = 20. We keep the same default hyperparameters across datasets to maintain a consistent audited comparison protocol.

Baseline implementations use scikit-learn (Pedregosa et al., 2011) for LR, SVM, NB, Random Forest, MLP, Ridge, and Random-Forest regression; reference implementations for XGBoost (Chen and Guestrin, 2016), LightGBM (Ke et al., 2017), and CatBoost (Prokhorenkova et al., 2018); and lightweight PyTorch reproductions for TabTransformer, FT-Transformer, and SAINT. Survey-style string variables are encoded as yes/no binaries, explicit gender categories, numeric range midpoints, or categorical codes depending on their observed value set.

Table 7: Leakage-audited task definitions.

Dataset	Classification label	Removed from input	Retained input groups
SMH-Kaggle	Academic risk: CGPA < 2.5	CGPA itself	Binary symptoms, help-seeking, demographics
MHP-Multi	PHQ-9 ≥ 10 or GAD-7 ≥ 10	PHQ-9 items, GAD-7 items, PHQ/GAD totals, current CGPA	PSS-10 items, demographics, scholarship/support proxy

Additional fairness diagnostics. The main paper reports the primary hyperparameter sweeps and ablation-style diagnostics. This appendix retains one supplementary view that is descriptive rather than central to the benchmark claim: a preliminary gender-sliced fairness comparison.

Table 8: Preliminary gender-sliced fairness diagnostic on leak-free MHP-Multi. The reference baseline is the strongest generic model from the audited MHP-Multi leaderboard (TabPFN v2). The "Prefer not to say" slice contains only 10 students and should not be over-interpreted.

Model	Subgroup	N	F_1	AUC	FNR
SHARE	Female	613	0.868	0.817	0.166
SHARE	Male	1405	0.803	0.810	0.274
SHARE	Prefer not to say	10	0.889	0.500	0.200
TabPFN v2	Female	613	0.924	0.839	0.012
TabPFN v2	Male	1405	0.891	0.831	0.042
TabPFN v2	Prefer not to say	10	1.000	0.500	0.000

Reproducibility. All experiments use fixed random seeds (1, 42, 2025). SMH-Kaggle is evaluated with 3-fold stratified cross-validation because the academic-risk task contains only six positive examples. MHP-Multi is evaluated with 5-fold stratified cross-validation. Preprocessing includes: (1) column normalization to a common schema; (2) yes/no, gender, and categorical encoding for survey responses; (3) range parsing for CGPA strings; and (4) feature exclusion according to the leakage audit in Table 7. In the ablation study, the $\alpha = 0$ setting disables the auxiliary GPA loss while retaining the shared prediction heads.

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